# UNIVERSITY OF UYO

DEPARTMENT OF MATHEMATICS

MTH 213 LINEAR ALGEBRA 1 SESSION 2025/2026

INSTRUCTION: ANSWER ANY FOUR QIESTION

TIME: 2 HRS

1a Let T: X → Y be defined as T $\{x_1, x_2, x_3\} = [3x_1 - 3x_2 + x_3, x_1 - x_2 + 2x_3, 3x_1 - 2x_2 + x_3]$. Determine the rank and nullity of T

1b Obtain the coordinate vector of S = [3,5,.2] relative to the bases X = $[u_1, u_2, u_3]$, where $u_1$ = [1,1,1], $u_2$ = [1,2,3] and $u_3$ = [1, -1,1]

2 Given the bases S and X of $R^3$, where S = [{1,2,2}, {0,1,1}, {1,0,1}] and X = [{1,2,3}, {3,1,2}, {2,1,1}]. Obtain the transition matrices A from S to X

3a Let T: $R^4 \to R^3$ be a linear transformation defined by T $(x_1, x_2, x_3, x_4) = (x_1 - x_2 + x_3 + x_4,\ 2x_1 - x_2 - x_3 + 2x_4,\ x_1 - 3x_2 + 2x_3 - x_4)$ Find the basis and dimension of the image of T

3b Express A = [ 3,4,5] as a linear combination of the vector X = [1,2,0], Y = [-1,1,3] and Z = [0,5,7]

4a Define [i] Rank and nullity of a linear mapping [ii] kernel and image of a linear mapping

4b Let V = $R^2$ with defined binary operations of vector addition and scalar multiplication. Show that V is a vector space

5a Define linear dependent and linear independent, investigate the linearity of the vector U = 2i + j – 3k, V = i – 4k and W = 4i + 3j – k

5b Define linear transformation and show that the mapping T: $R^2 \to R^2$ defined by T [x, y] = [ x + y, x] is linear

5c Find a linear map T: $R^4 \to R^4$ whose image is generated by [1, -1, 2, 0], [3, 1, -1, 3], [2,1,3, -1] and [ 1,1, -2,4]

6a consider the subspaces of R4 generated by X =[ [1,2,1,1], [0,1,1,1], [1,0,1,1] ] and Y = [ [1, -2,2,1], [1, -3,4,3], [1,2,3,4]. Find [i] dim [X + Y] [ii] dim [X n Y]

6b Define [i] Linear dependence [ii] linear combination [iii] basis